Determinants

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[This is essentially a transcribed version of the note written by Gowtham Kurri]
[I have only made minor editorial changes]

Story:

- Usually determinant is thought of
  as a quantity that can be computed for a matrix
  and is usually presented as a formula.

- Here, we take a different approach.
  We start by listing three desirable properties that we want
  a function $\det: F^{n\times n} \to F$ to satisfy.

  We postulate these as axioms and say that
  any function satisfying these axioms
  will be called *a* "determinant".

  Later, we argue that such a function is unique
  leading us to the well-defined formula for *the* determinant.

Definition. (Axioms for a Determinant)

Consider any function
$\det: F^{n\times n} \to F$
that we sometimes also denote by $|\cdot|$
where F is either R or C (real or complex).

We say $\det$ is a determinant if it satisfies the following properties:

(1) *Normalisation.*
$\det I = 1.$

(2) *Anti-symmetry.*
Let $A \in F^{n\times n}$

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The sign of the determinant $\det(A)$ flips

when any two rows of the matrix A are exchanged.

(3) *Linearity in each row.*

Determinant is linear in each row separately.

Let $A \in F^{n \times n}$ and denote by v_j the jth row of A, i.e.

$$A = \begin{bmatrix} \dots & v_1 & \dots \\ & \vdots & \\ \dots & v_n & \dots \end{bmatrix}.$$

Define the notation $\det(v_1, v_2, \dots v_n) \coloneqq \det(A)$.

Require that

$$\det(v_1, \dots, v_{k-1}, \quad c_1 v_k + c_2 u_k, \quad v_{k+1} \dots v_n)$$

$$= c_1 \det(v_1, \dots v_n) + c_2 \det(v_1, \dots u_k \dots v_n).$$

Story:

- Let us quickly recall the formulae for the determinant
  we learnt in school.

The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

- To ensure we understand this definition
  let us quickly verify on a few instances
  that these formuale
  satisfy all the properties listed above.

- Normalisation

$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ (set $b = c = 0$)

$\begin{vmatrix} 0 & 1 \end{vmatrix}$ (aa survives)

$\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} = 1$ (set $b=c=d=f=g=h=0$)
(only the aei term survives)

Antisymmetry:

$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ but $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$ ($-bc$ survives)

$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$ but $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = 1 \cdot (-1) = -1$

Linearity in each row:

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} \overset{\text{Exercise}}{=} \begin{vmatrix} a & \\ & \end{vmatrix}$$

Story:

- Note that

  $\det(A+B) \neq \det(A) + \det(B)$ in general;

  and similarly

  $\det(tA) \neq t\det(A)$ in general

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- In fact,
  it turns out that
  if A is an $n \times n$ matrix, then

  $\det(tA) = t^n \det(A)$.